25900112

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025
SECOND SEMESTER
Discipline Specific Core Course (DSC) MG2DSCHTC100 – PLANT PROPAGATION
(2024 Admission Onwards)

Duration: 1.5 Hours **Maximum Marks: 50**

Part A Short Answer

Answer any 10 questions

1. **Define vegetative mode of reproduction.**
   Multiplying or reproducing plants from vegetative parts like stem, root, or leaf without the use of seeds.
2. **Write any two importance of artificial method of plant propagation:**
   - It produces clones of parent plants.
   - Enables multiplication of plants that are unable to produce viable seeds.
3. **Which vitamin is essential for coenzyme formation in plant tissue culture?**
   Vitamin B1 (Thiamine).
4. **How does micropropagation differ from traditional plant propagation methods?**
   Micropropagation uses sterile lab set up and tissue culture techniques for rapid multiplication of plants whereas in traditional methods seeds or cuttings are used to propagate plants in soil.
5. **What is Wedge Grafting?**
   A type of grafting in which a V-shaped cut is made on the scion and a matching/complementary cut is made in the rootstock to ensure perfect fit.
6. **Mention the purpose of mist chamber in plant propagation.**
   Maintains high humidity to reduce transpiration and promote root development in cuttings. ***(Give marks for any other relevant uses)***
7. **Write an example for plant that is propagated through stem cutting:**
   Rose, Hibiscus, Bougainvillea etc.
8. **Define micropropagation:**
   Micropropagation is the *in vitro* propagation of plants from small plant parts called explants under aseptic conditions to produce a large number of genetically identical plants rapidly.

9. **Name the propagules that are propagated by division.**
   Rhizomes, tubers, bulbs, corms etc.

10. **Identify three essential utilities needed for a functional micropropagation laboratory:**
    - Laminar airflow cabinet
    - Autoclave
    - Growth chamber/ Incubator ***(Any)***

11. **What are offsets?**
    Type of vegetative propagule; Small daughter plants that form at the base of a parent plant that can be separated to grow independently.

12. **Why is layering used?**
    To induce root formation on a stem while still attached to the parent plant, ensuring better survival when planted in soil.

Part B Short Essay

Answer any 6 questions

13. **Why is gelling agent concentration important in micropropagation media?**
    Gelling agent influences medium firmness and water/nutrient availability. Low concentration of gelling agent in the medium can cause vitrification and very high concentration may inhibit root penetration.

14. **How does the nutritional state of parent plant impact rooting of cuttings?**
    Cuttings from healthy, carbohydrate-rich plants root more easily; high nitrogen reduces rooting success. ***(Liberal valuation)***

15. **Differentiate between dry heat and moist heat sterilization methods.**
    - **Dry heat**: Uses hot air ovens (160–180°C); ideal for glassware.
    - **Moist heat**: Uses autoclaves (121°C under pressure); ideal for media and tools.

16. **Explain the uses of different tools in plant propagation.**
    - **Grafting knife**: For grafting
    - **Secateurs**: For cutting plant parts
    - **Trowel**: For transplanting seedlings
    - **Watering can**: For watering
    - **Labels**: For plant identification ***(Any)***

17. **What are the key features of phytotrons?**
    Controlled chambers with regulated temperature, light, humidity, and CO_2, mainly used for plant research in simulated climates.

18. **Explain bulbs, pseudobulbs, corms and tubers with examples.**
    - **Bulbs**: Fleshy underground structures (e.g., onion).
    - **Pseudobulbs**: Swollen stems in orchids (e.g., Cattleya).
    - **Corms**: Solid stem bases (e.g., gladiolus).
    - **Tubers**: Swollen underground stems with eyes (e.g., potato).

19. **How is irrigation managed in a polyhouse, and what systems are most commonly used?**
    Irrigation is managed through automated drip or sprinkler systems. Fertigation systems are also commonly used to provide fertilizers along with irrigation. ***(Liberal valuation)***

20. **How do auxins influence rooting in plants?**
    Auxins stimulate root initiation by promoting cell division and elongation in the basal parts of cuttings.

Part C: Essay

Answer any 1 question

21. Describe the unorganized methods of plant micropropagation and explain their significance in plant tissue culture.

Unorganized methods involve callus formation from explants. Callus, the undifferentiated mass of cells, can be induced by specific hormonal conditions. The callus can then differentiate into shoots and roots.
Significance:

- Allows regeneration from a single cell.
- Useful in genetic and mutation studies.
- Production of disease-free plants. ***(Liberal valuation)***

22. Describe the factors influencing rooting in cuttings.

Rooting in cuttings is influenced by multiple factors:

- **Type of Cutting**: Softwood cuttings root faster than hardwood cuttings due to high metabolic activity.
- **Plant Hormones**: Auxins like IBA and NAA promote root formation.

- **Carbohydrates**: Serve as an energy source for new cell growth.
- **Environmental Factors**:
    - *Temperature*: Ideal range is 20–30°C.
    - *Humidity*: High humidity reduces water loss.
    - *Light*: Indirect light encourages rooting without stressing the cutting.
- **Parent Plant Health**: Juvenile, disease-free plants provide better cuttings.
- **Season**: Spring and fall are ideal for rooting due to moderate conditions.
- **Medium**: Should be porous, sterile, and retain moisture without waterlogging.
- **Additional Treatments**:
    - *Wounding* enhances hormone accumulation.
    - *Etiolation* and *girdling* improve rooting in difficult species.
    - *Fungicides* prevent infections during rooting. ***(or any other relevant factors)***